Smart Mug: ICUP

Group Digamma
LA 25

Introduction: Problem and Solution

- **Beverages are often too hot or cold**
  - Achieving the right temperature is a tedious process
  - Temperature changes too quickly
- **Bottle with Self-regulated temperature**
  - Preset by user
  - Set temperature sustained
  - Consists of heating and cooling elements

Fusion 360 CAD Model of Smart Bottle

- **Made with ABS (Lid) and PLA :**
  - Heat resistive material
  - 3D printable filament
- **Design Constraints:**
  - Has to fit a LCD display
  - Needs custom space for cooling / heating
  - Designed a custom battery pack at bottom (low center of gravity)
- **Design Specifications:**
  - Separate section for arduino internals
  - Design uses 3 M5x0.4 screws
  - Nylon filament withstands temps from -70 C to 250 C

cross section

¾ view

Design Parameters

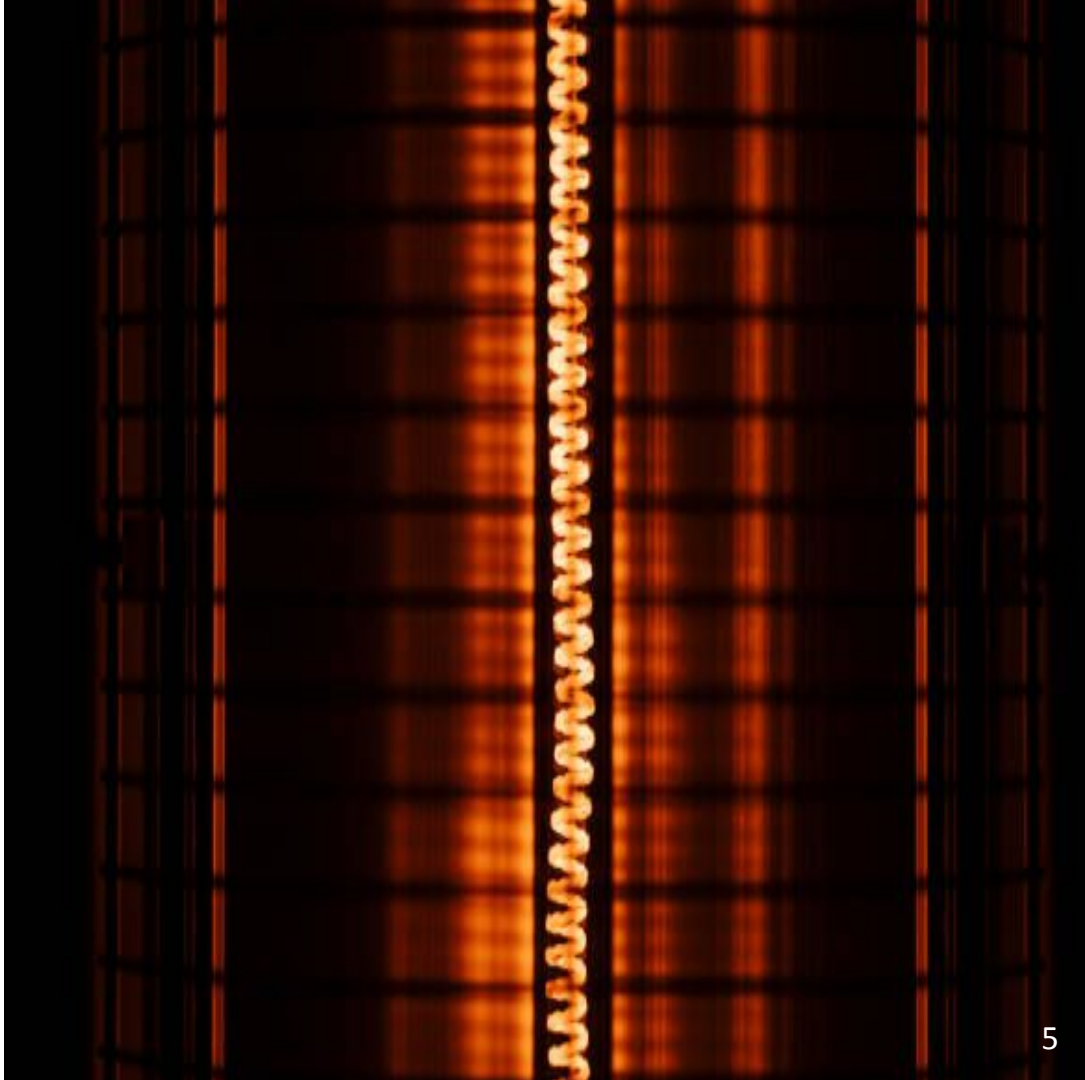
Finding balance between portability and efficiency, solving an optimization problem.

- Increase portability (smaller battery) sacrifices effectiveness (more time taken to reach max temperature differential, smaller temperature differential)
- Increase efficiency (heats faster and to higher temperature differential) sacrifices portability. (would need unfeasibly large battery
- Goal temperature differential is 35 C (25 C - 60 C)
- Finding a global maxima of power draw and efficiency was our engineering challenge

Heating

- **Past iterations**
  - Design flaws
  - inefficiencies
- **Current design**
  - 2nd law of thermodynamics
  - Composition of Aluminum
  - Static solution
- **Specifications of current design**

Specific Heat Capacity of Water:
4186 J / Kg



HEATING CALCULATIONS

Formula for Calculating Amount of Energy Required for Heating:

Q=mc∆t

Q = Total Energy (Joules)

m = mass (0.5kg)

c = Heat capacity of water (4.186 J/g degree C)

∆t = change in temperature (final - initial)

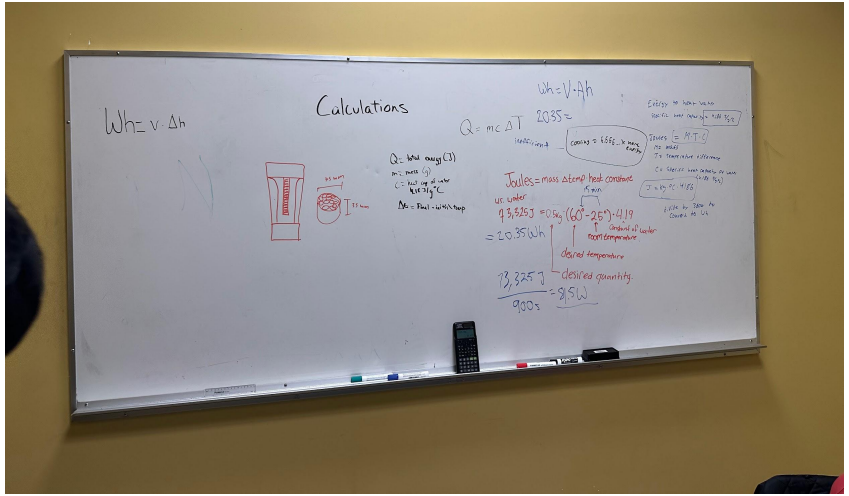
Final Desired Temperature = 60 Degrees

Initial Temperature = 25 Degrees (Room Temperature)

Maximum Time to Reach Desired Temperature: 15 minutes (900s)

Q = 0.5kg * 4.186 (60 - 25) = 73325 J

73325J / 900s = 81.5W



Cooling

-Cooling method (thermoelectric pads):

-Compact

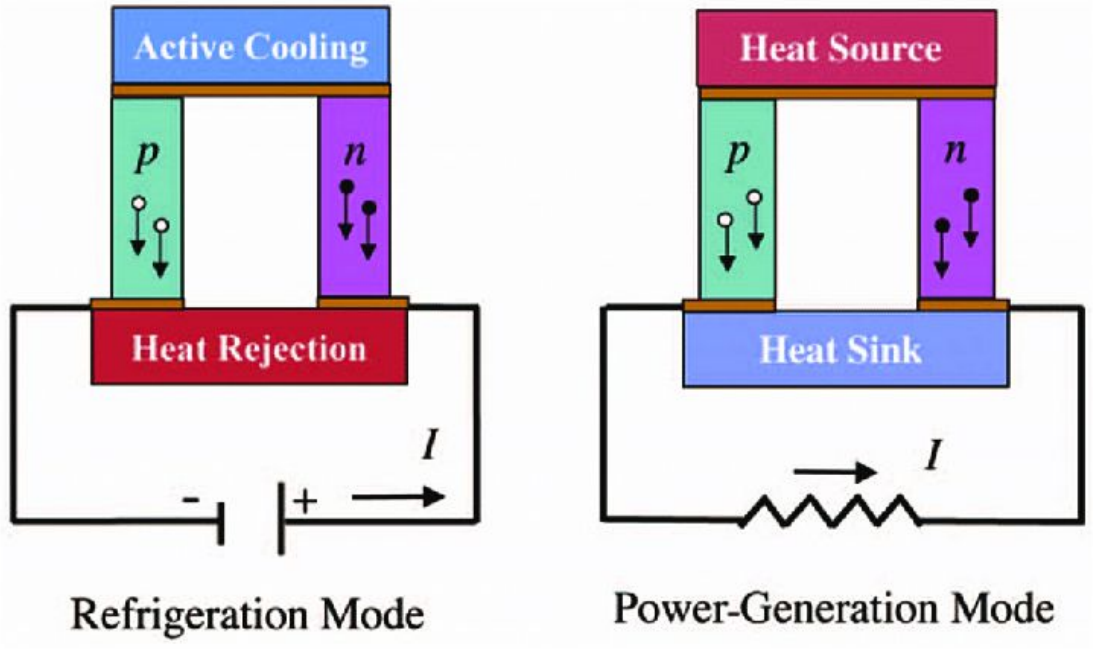
-Static

-Runs on electricity

-Sub-ambient temperature capabilities

Areas of concern:

-Energy efficiency



Power Supply

Design Parameters:

- At least 30Wh Capacity
- Li-ion battery chemistry
- Must fit within ~ 80 x 40 mm cylindrical bottom casing

Previous Design Considerations:

- AA batteries
- Off the shelf high power drone battery

Results

In order to build your desired pack, you need a 6 series, 20 amp BMS.

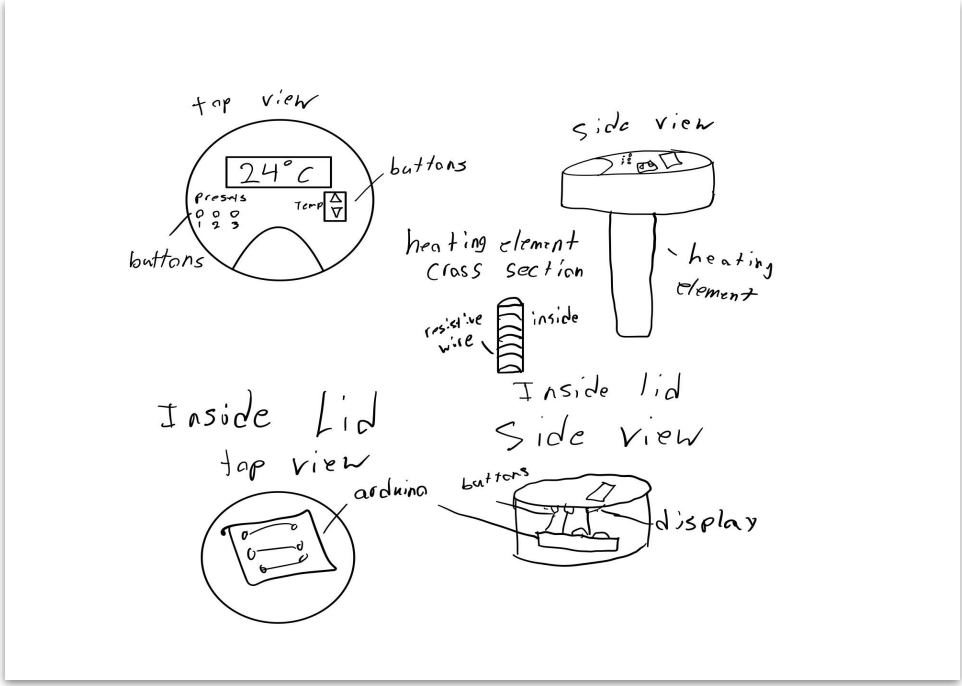
Nominal Pack Voltage	22.20 V
Minimum Pack Voltage	15.00 V
Maximum Pack Voltage	25.20 V
Max amps that the pack can generate	20.00 A
Max amps that the BMS needs to handle	14.80 A

Enter information on your planned pack in order to calculate which BMS you need.

Nominal voltage of one cell: 3.7	Maximum voltage of one cell: 4.2
Minimum voltage of one cell: 2.5	Capacity of one cell (Ah): 1.2
C Rate: 8.3333333	Series groups: 6
Cells in Parallel: 2	Maximum watt draw: 222

Integration & UI

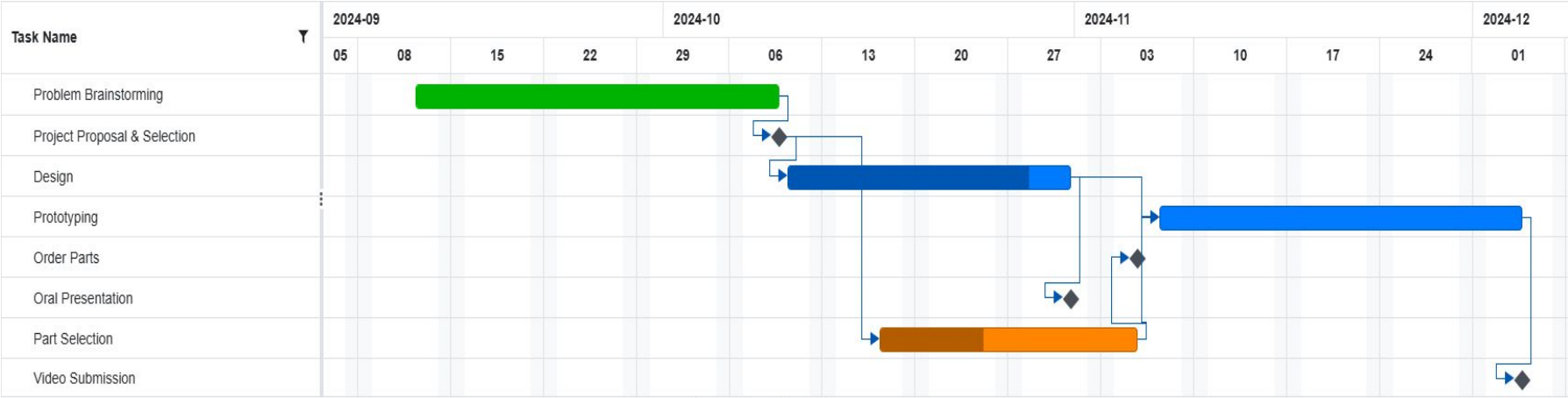
- The lid will have a UI to set temps
- Wires connect from the Lid UI to the power supply
- Up to 3 presets
- OLED screen with Membrane Buttons (water proof)
- We will be using Arduino Pro Mini Microcontroller
- Temperature Sensors in the cup



Conclusion

Name	Kalib	Yael	Taiyo	Ajay	Bhavan	Harjas	Amar
Brainstorming	Cooling & power	Heating power supply	Heating and parts needed	Cup design UI integration	Power and Cooling	Cup Design	Lid UI, Parts needed
Completed tasks	Designed cooling system and power source	Heating concept design	Heating concept design	3D CAD model	Designed battery case and cooling system	3D CAD Model	Designed lid
Ongoing tasks	Battery Selection & integration, wiring	Heating element construction	Heating element construction, wiring	Arduino integration, Construction and refining CAD Model	battery integration	Construction and refining CAD Model	UI element

Next Steps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2024-10
2024-11
2024-12
05
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27
03
10
17
24
01
Problem Brainstorming
Project Proposal & Selection
Design
Prototyping
Order Parts
Oral Presentation
Part Selection
Video Submission

Project Bill

Item	Price	Source
12V 50W Heating Rod (1)	$36.49	Amazon
MOSFET 30A 60V (10)	$11.99	Amazon
Arduino pro Mini (3)	$23.99	Amazon
5V regulator (5)	$12.55	Digikey
6S 22V Li-ion BMS (1)	$8.99	Amazon
Thermal Sensor (subject to change)	$10.80	Digikey
18350 Li-ion batteries (12)	$60 - $80	N/A

Total price: $165 - $185

Work Cited

Images:

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